ANDERSON JUNIOR COLLEGE
Preliminary Examinations 2009

MATHEMATICS
Higher 1 **8863/01**

Paper 1 16 September 2009
3 hours

Additional Materials:
Graph Paper
Formulae List

READ THESE INSTRUCTIONS FIRST

Answer **all** questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

When unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

The number of marks is given in [ ] at the end of each question or part question.

This Question Paper consists of **8** printed pages

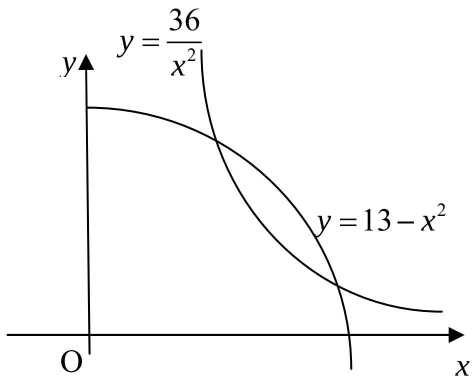
Section A: Pure Mathematics [40 marks]

1. The equation of a curve is $y = \ln(2x + a)$, where $a > 1$.

   Find the gradient of the curve at the point where the curve crosses the x-axis.

   Hence find the equation of the tangent to the curve at this point. You may leave your answers in terms of a where appropriate. [4]

2. The diagram below shows part of the graphs of $y = 13 - x^2$ and $y = \dfrac{36}{x^2}$ for $x \geq 0$.



   The two curves intersect at $(3, 4)$. Verify that the two curves also meet at $(2, 9)$. [1]

   Use integration to find the area of the region between the two graphs, showing your steps clearly. [4]

3. The functions f and g are defined by

$$\mathrm{f} : x \mapsto e^{2x} - 1, \quad x \leq 0,$$

$$\mathrm{g} : x \mapsto \frac{1}{x+1}, \quad x > -1.$$

   (i) Explain why f has an inverse function.
   Find $\mathrm{f}^{-1}(x)$. State the domain of f^{-1}. [3]

   (ii) The composite function gg exists for $x > -1$.
   Write down $\mathrm{gg}(x)$, simplifying your answer.
   Find the range of gg. [3]

   (iii) Find $(\mathrm{gg})^{-1}(x)$. [1]

4. (i) Sketch the graph of $y = \tan x$ for $0^0 < x < 540^0$, showing clearly all axial intercepts and asymptotes. [2]

(ii) Insert in your sketch, the graph of $y = \cos x$ for $0^0 < x < 540^0$. [1]

The smallest positive root of $\tan x = \cos x$ is denoted by α.
State in terms of α, a root of $\tan x = \cos x$ in the range

(a) $90^0 < x < 180^0$. [1]

(b) $360^0 < x < 540^0$. [1]

State the number of roots for $\tan x = \cos x$ in the range $0^0 < x < 360n^0$. [1]

5. (a) A curve has an equation $y = a^3 x + \dfrac{4}{x^2}$, where a is a positive constant.

Find an expression for $\dfrac{\mathrm{d}y}{\mathrm{d}x}$, leaving your answer in terms of a. [1]

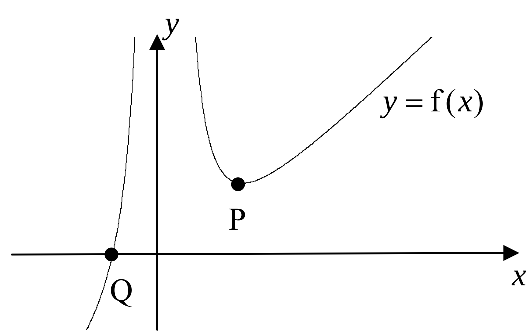
Show that the curve has a stationary point when $x = \dfrac{2}{a}$.

Find the y-coordinate of this stationary point, leaving your answer in terms of a. [2]

Briefly explain how you can tell that the curve is increasing when $x > \dfrac{2}{a}$. [1]

(b) The following shows the graph of $y = \mathrm{f}(x)$, where $\mathrm{f}(x) = a^3 x + \dfrac{4}{x^2}$.

P is the minimum point with x-coordinate $x = \dfrac{2}{a}$. The curve cuts the x-axis at $\mathrm{Q}\left(\dfrac{-1.59}{a}, 0\right)$.



(i) Sketch the graph of $y = \mathrm{f}'(x)$, showing clearly where this curve cuts the x-axis. [2]

(ii) Sketch on a separate diagram, the graph of $y = |\mathrm{f}(x)|$, showing clearly the points corresponding to P and Q.
State the range of values of x for which $\mathrm{f}(x) = |\mathrm{f}(x)|$. [2]

6. (a) Find in exact form, the values of m for which $(3m-2x)^2 > 10 - x^2$ for all real values of x. [3]

(b) (i) If k is a positive constant, find the x co-ordinates of the points where the line $y = k - 2x$ meets the curve $y = \dfrac{k^2}{k-2x}$, leaving your answers in terms of k where appropriate. [1]

(ii) Consider the case where $k = 3$.
Sketch the graph of $y = \dfrac{9}{3-2x}$, showing clearly any asymptotes and the point(s) where the curve cuts the axes.

By inserting an additional graph in your diagram, or otherwise, find the values of x for which $\dfrac{9}{3-2x} \le 3 - 2x$. [3]

(iii) Deduce in exact form, the values of x for which $\dfrac{9}{3-2(x-2)^2} \le 3 - 2(x-2)^2$. [3]

Section B: Statistics [40 marks]

7. A man drives to work and he may be delayed at two road junctions, X and Y. If he is delayed at either junction, he will be late for work.

The probability that he will be delayed at the first junction X is $\frac{1}{10}$.

The probability of him being delayed at the second junction Y is as follows:

- It is twice the probability of being delayed at the first junction X if he has been delayed at the first junction X
- It is the same as the probability of being delayed at the first junction X if he is not delayed at the first junction X.

(i) Find the probability that the man is late for work on a randomly selected day. [2]

(ii) Given that the man is late for work on a particular day, find the probability that he is delayed at the first junction X. [2]

(iii) If the man has been delayed at only one junction, find the probability that he was delayed at the first junction X. [2]

(iv) The man works from Monday to Friday. Find the probability that in a particular week, he is late for work on Monday and Friday, but on time for the other three days. [2]

8. A confectionery makes a large number of lollipops each day, of which 20% are strawberry lollipops, 30% are mango lollipops, and the rest are a mixture of other types of fruit lollipops.

(i) Grace picks 12 lollipops at random.

(a) Find the probability that she has exactly 3 strawberry lollipops. [1]

(b) Find the probability that she has more than 3 strawberry lollipops. [2]

(c) If it is known that the first lollipop that Grace picks is a strawberry lollipop, find the probability that Grace has exactly four strawberry lollipops in total. [2]

(ii) A candy shop owner buys 100 lollipops at random. Use a suitable approximation to find the probability that more than 25 are mango lollipops. [3]

A packet contains 12 lollipops packed at random.

(iii) Ann buys 20 packets of lollipops at random to sell at her college fun fair.
Find the probability that she has bought at least 3 packets, but less than 8 packets that contain exactly 3 strawberry lollipops. [3]

9. (a) Aspire Junior College has 2500 students. The profile is shown in the following table.

	Year	
	One	Two
Girls	700	650
Boys	550	600

The Library and Computer Clubs in the college intend to choose a sample of 100 students to find out the weekly pocket money of the students.

The Library Club wants to use systematic sampling. Explain how they use systematic sampling to select the sample of 100 students. [2]

The Computer Club argues that stratified sampling will give more representative results. Explain, with reasons, whether you agree with them, and describe how they can carry out the stratified sampling to select the sample of 100 students. [3]

(b) It is known that in the nation, the weekly pocket money of a junior college student is normally distributed, with a mean of μ dollars, and a standard deviation of 4 dollars.

(i) If 70% of the junior college students receive at least 25 dollars in a week, find the value of μ, correct to 3 significant figures. [2]

Take the value of μ as 27.1 in the following parts of the question.

(ii) A group of 10 junior college students from various junior colleges are randomly chosen. Find the probability that the average weekly pocket money for this group is more than 30 dollars. [2]

(iii) Explain, with reasons, whether it is necessary that the weekly pocket money of a junior college student follows a normal distribution for the probability in (ii) to be valid. [1]

10. A fruit stall sells 3 types of durians: D24, Sultan and XO.
For each type of durian, a randomly chosen durian is normally distributed with mean and standard deviation as shown in the following table:

	Mean (kg)	Standard deviation (kg)	Price per kg
XO	2.5	0.8	$7
Sultan	2.0	0.4	$4
D24	2.0	0.5	$4

Mr Lee picks two XO durians, and Mrs Lee picks one Sultan and three D24 durians at random.

(i) Find the probability that none of the four durians picked by Mrs Lee weighs less than 2.4 kg. [3]

(ii) The probability that the total weight (in kg) of the four durians chosen by Mrs Lee lies in the range (7.5, a) is 0.4. Find the value of a. [3]

(iii) Find the probability that the total cost of the two XO durians that Mr Lee picks is more than the total cost of the four durians that Mrs Lee picks. [3]

(iv) If the stallholder gives a discount of 50 cents for each durian, state the mean and the standard deviation of the total cost of the six durians after the discount. [2]

11. In a particular chocolate factory, its premium chocolates are packed in 250 g packets. The weight of the packets is normally distributed and has a standard deviation of 10 g.

The quality control engineer claims that the packing machine is not working properly and that the mean weight of the packets of premium chocolates is not 250 g.
He takes a random sample of 60 packets of chocolates and notes down the weight of chocolates (x g) of each packet.
The data is summarized as $\sum x = 14898$ and $\sum (x - 248.3)^2 = 1980$.

(i) Find the unbiased estimates of the population mean and variance of x. [2]

(ii) Test, at 4% significance level, whether the quality control manager is justified in his claim? [3]

Explain what is meant by the phrase “at 4% significance level” in the context of this question. [1]

Meanwhile, the sales manager suspects that the mean weight of the packets of premium chocolates is less than 250 g. He carries out another test. He weighs another 60 packets of chocolates. He obtains a sample mean weight of k g, which gives a p-value of 0.04. State the value of k. [2]

12. The human resource manager in a departmental store would like to find out the relationship between the length of experience of a sales personnel and his ability to bring in sales.

Eight sales personnel from his department store are selected at random. The length of the person's experience in the company, x months, and his monthly sales revenue, y thousand dollars, are recorded.

x (months)	10	14	18	22	30	32	34
y (in thousand dollars)	15	18	23	29	39	45	53

(i) Draw a scatter diagram, and explain clearly what you can deduce about the relationship between the length of a sales personnel's experience and his sales revenue. [2]

(ii) Obtain the equation of the regression line of y on x in the form $y = ax + b$.
Explain what the value of a means in the context of this question. [2]

Obtain the equation of the regression line of x on y. [1]
State the co-ordinates of the point where the two regression lines will meet. [1]
Insert the regression line of y on x on your scatter diagram. [1]

(iii) Obtain the product moment correlation coefficient. [1]

(iv) Use the appropriate regression line to estimate the monthly sales revenue of a salesman with exactly two years experience in the company.
Comment on the reliability of your estimate. [2]

(v) If the length of experience of the sales personnel is measured in years instead, explain whether it would make a difference in the value of the product moment correlation coefficient. [1]

The human resource manager of another department store hears of this practice, and uses the regression line of y on x in (ii) to estimate the earnings of his workers for his department store.
Comment on the reliability of using this regression line. [1]

End of Paper